Annotations Draft - G8906 Craft and Science

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p145r: Cuttlefish bone

Do not cast tin or lead too hot, because it would burn the bone & become lumpy. And to know when it is the right temperature, dip a little piece of paper in it. If it turns black without catching fire, it is the right temperature. But if it burns & makes a fire, it is too hot. Gold & silver can indeed be cast, but it never comes out clean. To cast something delicate well, it is necessary that the bone not be extremely dry, because it is rough & does not strip neatly & crumbles & scales off. In any case, before you cast, dry them [the bones] & principally for gold, which does not admit any humidity. You will know that they are dry enough when, after putting them near some fire for a bit, they cry & crackle inside & when you press them when you bring your ear close. Now join the bones & daub the joints with a little clay and slowly dry it on a fire & cast & then shake the mold or scratch the rough crannies and let it cool down before opening it. Commonly one cuts the bone in the middle, and the dull part that does not have half circles is more delicate and smoother to mold, and therefore on always presses it first, the other is scaled on the inside just as is manifest on the outside. Even so one is not helped at all by this with delicate things that are molded in two places. Make smooth & flatten these two halves on some even piece of wood, then grate some charcoal on top make it strip well. And to make the charcoal run evenly everywhere, hit the side which holds half the bone with your hand. Being both covered in charcoal, take the first bone that was prepared & rounded on the edges, and having posed the medal above it, thrust it down hard and press very hard. But with fingers you cannot press evenly, because if you press on the edges, the middle will remain hollow. Start therefore with the middle, & then do the edges. But to do it better, put something flat & even on the medal, or some large square file, & press with that; because you will press evenly, sometimes with your knee, other times with your foot with your shoe taken off. & do it from above, so that you impress it without breaking it. If it is not well molded on the first go, do it again several times. Then restore and flatten your halves of molded bone from the sides. And to cut them well, always start from the most tractable part, coming out to the shell. And if your medal does not come out on its own, scratch the rough bone from behind and it will strip. When you want to cast, settle & affix the two bones with little pins of wood & in order to cast neatly, you have only to try it with sand.

Because we had unique access to gold and a gold casting facility (and Dr. Boyd, an expert maker of sorts[1], to work with), we chose to work with two recipes relating to

[1] While it may seem unusual to refer to a dentist as an expert maker of cast gold objects, in fact Dr. Boyd has been using modern techniques to cast gold crowns, fillings, inlays and onlays for teeth for 50 years. Although today it is most common for dentists to outsource this type of laboratory work, Drs. Boyd, PC. is one of a handful of offices in New York City where the doctors do their own work in gold in order to insure quality.

gold casting. We chose this recipe, along with 106r, because we felt they could be assayed easily (and fruitfully) in combination.

Recipe 145r, “Cuttlefish bone,” is a recipe for how to use cuttlefish bone as a mold for casting in metal. Although there are several recipes in the manuscript for casting in cuttle molds, we chose this one because it refers specifically to casting *gold* in cuttlefish. This recipe describes how to know when certain metals are the right temperature to cast, what metals work best in cuttle molds, how to prepare the cuttle molds for impression, how to create the impression, and how to bind the mold to prepare for casting.

There are four recipes in the manuscript that deal with cuttlefish molds.[2] Of these recipes, the most significant for our purposes (aside from 145r) was 072v, a general recipe on casting that includes information about cuttlefish. 072v states, “Cuttlefish bone molds lead better than [any]thing there is...” As we shall explain, this piece of information proved invaluable for understanding the significance of a line in 145r, “Gold and silver can indeed be cast, but it never comes out clean.”

It is not unusal that the maker did not describe too many experiments with cuttle molds: other contemporary references to cuttlefish casting we found simply gloss over how to do it. Cennino Cennini mentions cuttlebone in recipe about something else, stating, “…Rubbed and smoothed down with cuttle such as the goldsmiths use for casting…”[3] and likewise Vannoccio Biringuccio makes a passing reference to cuttle

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[2] See 072v, 091r, 139r, and 157r.

[3] Cennini, Cennino, and Daniel V. Thompson. *The Craftsman's Handbook*. New York: Dover Pubications, 1954. Print.

casting in the *Pirotechnia*, “To save time, they cast all metals in cuttlebone, having first molded in halves the object they are to make.”[4] It seems the practice was so widely known that the methods were obvious to contemporary makers and they did not feel the need to comment about them extensively.[5]

The author may not have been very interested in recording cuttlebone casting recipes, but he was certainly extremely interested in investigating best practices for casting in gold. The manuscript is full of gold casting recipes, many suggesting the use of “spalt” “sphalt” or “asphalt” as the optimal medium for creating a mold in which to cast gold.[6] In fact, as we have mentioned above, our recipe explicitly states that cuttlebone is *not* the best medium for casting in gold. We elected to attempt gold casting in cuttlebone to ascertain why that might be, as it seems like a medium that is extremely easy to mold.

The major question that immediately arose for us upon first consulting this recipe was: what is the meaning of “never comes out clean” in reference to casting gold and silver in cuttlebone? Does this mean the cast object doesn’t come out easily from the mold (and therefore the mold cannot be reused)? Or does it mean it does not take a good impression and therefore is not a good medium for casting gold or silver (this seems unlikely as this method is still used today)? We were also very interested in exploring the different bodily practices the maker describes for pressing the object into the mold (hands, knees, feet), and to ascertain whether pressing with less conventional body parts

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[4] Biringucci, Vannoccio. *The Pirotechnia of Vannoccio Biringuccio: The Classic Sixteenth-century Treatise on Metals and Metallurgy*. New York: Dover Publications, 1990. Print.

[5] Our class encountered a very similar obstacle when researching bread recipes: it seems the knowledge of how to make bread was so common to early modern readers of recipe books that it was difficult to find a recipe for it.

[6] See 015r, 047r, 047v, 106r, 106v, 107r, 108r, 116r, 116v, 117v, 119v, 120r, 124v, 134v, 135v, 138v, 145r, 156r, 156v, and 157r.

made any kind of difference in the outcome of the mold. We will address our findings as we walk through our process for following the recipe.

We began with cuttlebones. We first needed to prepare the bones to be molded. The maker's first instruction was to dry the bones by a fire until they "cry and crackle" when you press them. Our bones were already quite dry when we acquired them, but since it is impossible to know in what state the maker would have acquired his cuttlebones[7], we followed the instruction. Although we did not have access to a fire, we held the cuttlebones over an electric burner using a thin mesh strainer. Much to our delight, they really did emit a noise that sounded much like a "cry and crackle" after a few minutes, without even needing to be pressed.

Next, we bisected them and sanded them down so that they had a flat surface for the molding. One question that arose was what the recipe means by, "commonly one cuts them in half." We were not sure whether the maker meant transversely or sagitally, and some bones are even thick enough that it would be possible to bisect them coronally[8]. We decided it made the most sense in terms of the function of the mold to bisect horizontally, and used a jeweller's saw to cut them. This was exceptionally slow going as the bones were brittle and could break easily. However it became easier with successive attempts as we became more familiar with resistance of the materials. We cut a series of molds between one and three inches in height (using one half of each cuttle for the front and back of a given mold), with flat bottoms so that they could rest without being held during the pouring. We then sanded these down ("Make smooth and flatten these two halves on

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[7] For example, he may have been gathering his own cuttlebones directly from the seaside, in which case they may have arrived at his laboratory quite damp.

[8] Indeed, Dr. Boyd made a sample mold from the same instruction but without watching the videos, and while we felt it made the most sense to bisect horizontally, Dr. Boyd bisected his bone coronally.

some even piece of wood," the recipe states) simply by rubbing the softer side of the cuttlebone against a plywood board until they were flat.

Making the impression was incredibly easy. As with the cutting, at first we had some failed attempts because the material is quite brittle. But as we developed a deeper understanding of its resistance, we were able to create a mold of a ring simply by pressing it into the bone. We also molded a small cross shape with bulbous ends, which Dr. Boyd provided. Our aim in pouring these two shapes, as expressed in our annotation for 106r, was to test the success of our flux by pouring shapes that might prove an obstacle for flow. The recipe is correct that it is difficult to press evenly using just your fingers. We tried pressing with our bare feet, per the recipe's suggestion, and did not find this a helpful alternative, breaking several molds in the process as it was difficult to be precise with our pressure. Eventually we found that the best way was to get the impression started in the right spot with our fingers on one side of the mold, then to sandwich the ring in between the two sides, sit in a chair, and use our palms in between our knees so that our knees could apply strong and even pressure, but our palms could guide and moderate it. This was the method we developed, based on the maker's guidance, of quickly taking the most successful impression. We also used a razor to cut a funnel-shaped pouring cup in the top of the mold for the metal to enter, and three or four small, one-sided vents.

To fasten the two halves of the mold together we used small clippings of wire in place of the wooden pins the maker recommends.[9] Time permitting we would have liked to attempt daubing the joints of the mold shut with the clay per the recipe's instructions.

[9] We did not feel this would impede historical accuracy because the function of the pins remains exactly the same.

However, we were wary of miscalculating the clay's dryness and pouring gold into a mold that was not fully dry, which could create steam and make the gold splatter.[10] In the interest of safety we used a modern alternative, binding wire.

Next we attempted to pour our molds. After several unsuccessful attempts with beeswax (the mold was too porous and it just absorbed the wax) and sulfur (the sulfur was too brittle and could not be removed in one piece), we poured our first tin. Since we are not working with pouring lead in this workshop, we felt tin was the safest possible alternative because of their similar melting points. This could perhaps tell us about why the maker felt cuttlebone was such a perfect medium for lead casting, in order to better understand why the maker felt it was a less than optimal metal for gold casting. The tin trials were extremely successful. After it cooled, we were able to pop the tin ring directly out of the mold, in one piece, and without damaging the mold at all. The melting point of tins is so low that the mold was not even slightly discolored, much less damaged in any way. In fact, it was so perfectly in tact that we were able to cast an identical ring from a second pour into the same exact mold. It immediately became clear to us what the maker meant for something to "come out clean" from a mold. It seemed likely that because the melting point of gold is so much higher (nearly four times the melting point of tin), the gold would burn the mold, destroying it and rendering it useless for future trials.

Our next attempt was at pouring gold. As discussed in our annotation on 106r, "Making gold run for casting," our verdigris flux functioned exceedingly well for cleaning the gold and increasing its flow. The first mold we poured was Dr. Boyd's impression of the small cross. The casting went perfectly, and the cross came out beautifully. However, as we suspected the inside of the mold was completely destroyed.

[10] This occurred with our sandcasting molds when we tried to cast silver at Ubaldo Vitali's workshop.

The molten gold burnt and blackened it, so that when the gold casting was removed, the mold flaked and crumbled, and as the maker predicted, it did not come out clean. We next made three attempts at pouring the gold for the ring. The pouring was somewhat rockier in each of these attempts, and we could not get the gold to flow through the whole mold. Dr. Boyd suggested that perhaps enlarging our sprue would help by making it easier for us to pour quickly and accurately, and we also expanded the bottom of the funnel on each side so the gold would be able to flow more easily in both directions around the ring. After a second try using these same procedures, we were able to create a near perfect casting of the ring in gold.

Having now performed a series of experiments with cuttlebone casting, it is obvious why the maker preferred using cuttle for metals with a lower melting point: there is a savings of time and materials if one is able to mold something once and then create several castings from the same mold before it begins to deteriorate. It is also clear why the maker wrote that cuttlebones do work for gold and silver, but less well, because we achieved a very successful casting although we destroyed our molds in the process. However, our success may in part be due to other factors than the mold itself: for example, the propane torch with which we melted the gold was very hot and perhaps more effective than early modern heating methods at making the gold run. Another factor that may have made the early modern maker wary of cuttle casting is the pattern of ridges that the bone leaves on everything that is cast in it. Since the recipe makes clear that the maker was casting *medals* in cuttlebone, perhaps cuttlebone was unappealing to him because of these marks, which may have detracted from the intended appearance of the medal's surface. However, when our items were firmly pressed into the bone, these marks were far less significant than we had imagined they would be. We were also working with very basic shapes that laxzcked detail. Perhaps cuttlebone casting was of less interest to the maker because the figures he was working with were more intricate than our simple objects. Nonetheless, our casting was so successful that it does lead us to wonder whether our modern tools led to a grossly different outcome than the early modern equivalents, or else to question why the maker did not recommend cuttlebone casting more highly.